

Problem 284: To Catch a Satellite

Difficulty: Easy

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Originally Published: Code Quest 2026

Problem Background

There are thousands of satellites spinning around over the Earth, but they're not the only things flying around in space around us. There's also tiny meteors, junk cast off by astronauts, and bits of damaged or destroyed satellites flying around as well. All of this space debris presents a potential danger to the still-functioning satellites, and any impacts can cause serious damage.

As part of your internship with Lockheed Martin Space, you've been asked to create a program that can calculate certain properties of a satellite, so missions can be planned to go out and repair the satellite when needed.

Problem Description

Satellites all move in high orbit above the earth, and their altitude is closely related to both their velocity (how fast they're going) and their period (the amount of time it takes to complete one circuit around the Earth). We need to know both of these properties when planning how to send astronauts or robots to repair the satellite.

In order to calculate these values, you'll need to know the altitude of the satellite (above Earth's sea level) and the following formulas and values:

- Universal gravitational constant (Nm^2/kg^2): $G = 6.673 \times 10^{-11}$
- Mass of the Earth (kilograms): $M = 5.98 \times 10^{24}$
- Radius of the Earth (meters): $R_{\text{Earth}} = 6.37 \times 10^6$
- Satellite orbital radius (meters): $R = R_{\text{Earth}} + H_{\text{satellite}}$
- Satellite orbital velocity (meters per second):

$$V = \sqrt{\frac{G * M}{R}}$$

- Satellite orbital period (seconds):

$$T = \sqrt{\frac{4 * \pi^2 * R^3}{G * M}}$$

Sample Input

The first line of your program's input, **received from the standard input channel**, will contain a positive integer representing the number of test cases. Each test case will include a single line containing a positive integer, representing the altitude of the satellite in meters above Earth's sea level ($H_{\text{satellite}}$).

```
3
35786000
1900000
200000
```

Sample Output

For each test case, your program must print a single line containing the following information about the satellite, separated by a space:

- The orbital velocity of the satellite, in meters per second, and
- The orbital period of the satellite, in H:MM:SS format. Include leading zeroes as needed to ensure exactly two digits are printed for minutes and seconds, and at least one digit is printed for hours (e.g. a period of 0 seconds should be printed as 0:00:00)

For both values, round your results to the nearest integer using the rounding rules provided in the Reference Materials.

```
3077 23:54:51
6946 2:04:40
7793 1:28:17
```